Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A thick uniform rubber rope of density $1.5\ \mathrm{g\ cm^{-3}}$ and Young's modulus $5 \times 10^{6}\mathrm{Nm^{-2}}$ has a length of $8\ \mathrm{m}$. When hung from the ceiling of a room, the increase in length of the rope due to its own weight will be

(A) $9.6 \times 10^{-2}\ \mathrm{m}$

(B) $19.2 \times 10^{-3}\ \mathrm{m}$

(C) $9.6 \times 10^{-3}\ \mathrm{m}$

(D) $9.6\ \mathrm{m}$

Q.2

Two wires A and B of the same material have their lengths in the ratio of $1:2$ and their diameters in the ratio of $2:1$. If they are stretched with the same force, the ratio of the increase in the length of A to that of B will be

(A) $1:2$

(B) $4:1$

(C) $1:8$

(D) $1:4$

Q.3

A metal bar of length L and area of cross-section A is rigidly clamped between two walls. The Young's modulus of its material is Y and the coefficient of linear expansion is α. The bar is heated so that its temperature increases by $\theta^{\circ}\mathrm{C}$. Then the force exerted at the ends of the bar is given by

(A) $YL\alpha\theta$

(B) $YL\alpha\theta/A$

(C) $YA\alpha\theta$

(D) $Y\theta\alpha/LA$

Q.4

A wire of length L is stretched by a length l when a force F is applied at one end. If the elastic limit is not exceeded, the amount of energy stored in the wire is given by

(A) $F \times l$

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(B) $1/2(F \times l)$

(C) Fl^2/L

(D) $1/2Fl^2/L$

Q.5

A wire, suspended vertically from one end, is stretched by attaching a weight of $20\ \mathrm{N}$ to the lower end. The weight stretches the wire by $1\ \mathrm{mm}$. How much energy is gained by the wire?

(A) $0.01\ \mathrm{J}$

(B) $0.02\ \mathrm{J}$

(C) $0.04\ \mathrm{J}$

(D) $1.0\ \mathrm{J}$

Q.6

A uniform steel wire of length $4\ \mathrm{m}$ and area of crosssection $3 \times 10^{-6}\ \mathrm{m}^2$ is extended by $1\ \mathrm{mm}$ by the application of a force. If the Young's modulus of steel is $2.0 \times 10^{11}\mathrm{Nm}^{-2}$ the energy stored in the wire is

(A) $0.025\ \mathrm{J}$

(B) $0.050\ \mathrm{J}$

(C) $0.075\ \mathrm{J}$

(D) $0.100\ \mathrm{J}$

Q.7

The following four wires are made of the same material. Which of these will have the largest extension when the same tension is applied to each?

(A) length $= 0.5\ \mathrm{m}$, diameter $= 0.5\ \mathrm{mm}$

(B) length $= 1.0\ \mathrm{m}$, diameter $= 1.0\ \mathrm{mm}$

(C) length $= 2.0\ \mathrm{m}$, diameter $= 2.0\ \mathrm{mm}$

(D) length $= 3.0\ \mathrm{m}$, diameter $= 3.0\ \mathrm{mm}$

Q.8

A steel wire of diameter $2\ \mathrm{mm}$ has a breaking strength of $4 \times 10^5\ \mathrm{N}$. The breaking strength of similar steel wire of diameter $1\ \mathrm{mm}$ will be

(A) $4 \times 10^5\ \mathrm{N}$

(B) $2 \times 10^5\ \mathrm{N}$

(C) $1 \times 10^5\ \mathrm{N}$

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(D) 0.5×10^5 N

Q.9

A copper wire of length 1.0 m and a steel wire of length 0.5 m having equal cross-sectional areas are joined end to end. The composite wire is stretched by a certain load which stretches the copper wire by 1 mm. If the Young's modulii of copper and steel are $1.0 \times 10^{11}\mathrm{Nm}^{-2}$ and $2.0 \times 10^{11}\mathrm{Nm}^{-2}$, the total extension of the composite wire is

(A) 1.25 mm

(B) 1.50 mm

(C) 1.75 mm

(D) 2.0 mm

Q.10

Two springs of equallengths and equal cross-sectional areas are made of materials whose Young's modulii are in the ratio of $3:2$. They are suspended and loaded with the same mass. When stretched and released, they will oscillate with time periods in the ratio of

(A) $\sqrt{3}:\sqrt{2}$

(B) $3:2$

(C) $3\sqrt{3}:2\sqrt{2}$

(D) $9:4$

Q.11

One end of a uniform rod of mass M and crosssectional area A is suspended from rigid support and an equal mass M is suspended from the other end. The stress at the mid-point of the rod will be

(A) $\frac{2Mg}{A}$

(B) $\frac{3Mg}{2A}$

(C) $\frac{Mg}{A}$

(D) zero

Q.12

Two rods of different materials, having coefficients of linear expansion α_1 and α_2 and Young's modulii Y_1 and Y_2, are fixed between two rigid walls. The rods are heated to the same temperature. There is no bending of rods. If $\alpha_1 : \alpha_2 = 2:3$, the thermal stresses developed in the two rods will be equal provided $Y_1 : Y_2$ is equal to

(A) $2:3$

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(B) $1:1$

(C) $3:2$

(D) $4:9$

Q.13

A uniform wire of cross-sectional area A and Young's modulus Y is stretched within the elastic limit. If S is the stress in the wire, the elastic energy density stored in the wire in terms of the given parameters is

(A) $\frac{S}{2Y}$

(B) $\frac{2Y}{S^2}$

(C) $\frac{S^2}{2Y}$

(D) $\frac{S^2}{Y}$

Q.14

One end of a uniform wire of length L and of weight W' is attached rigidly to a point in the ceiling and a weight W_1 is suspended from its lower end. If S is the area of cross-section of the wire, the stress in the wire at a height $3L/4$ from its lower end is

(A) W_1/S

(B) $\left(W_1+\frac{W}{4}\right)/S$

(C) $\left(W_1+\frac{3W}{4}\right)/S$

(D) $\left(W_1+W\right)/S$

Q.15

A uniform wire (Young's modulus $2\times10^{11}\mathrm{Nm}^{-2}$) is subjected to a longitudinal tensile stress of $5\times10^{7}\mathrm{Nm}^{-2}$. If the overall volume change in the wire is 0.02%, the fractional decrease in the radius of the wire is

(A) 1.5×10^{-4}

(B) 1.0×10^{-4}

(C) 0.5×10^{-4}

(D) 0.25×10^{-4}

Q.16

An elastic spring of unstretched length L and force constant k is stretched by a small length x. It is further stretched by another small length y. The work done in the second stretching is

(A) $\frac{1}{2}ky^2$

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(B) $\frac{1}{2}k\left(x^2+y^2\right)$

(C) $\frac{1}{2}k(x+y)^2$

(D) $\frac{1}{2}ky(2x+y)$

Q.17

The Poisson's ratio of a material is 0.4. If a force is applied to a wire of this material, there is a decrease of cross-sectional area by 2%. The percentage increase in its length is:

(A) 3%

(B) 2.5%

(C) 1%

(D) 0.5%

Q.18

A stone of mass m is attached to one end of a wire of cross-sectional area A and Young's modulus Y. The stone is revolved in a horizontal circle at a speed such that the wire makes an angle θ with the vertical. The strain produced in the wire will be.

(A) $\frac{mg\cos\theta}{AY}$

(B) $\frac{mg}{AY\cos\theta}$

(C) $\frac{mg\sin\theta}{AY}$

(D) $\frac{mg}{AY\sin\theta}$

Q.19

A block of mass M is suspended from a wire of mass m, cross-sectional area A and length L. If all the energy stored in the wire is converted into heat, the rise in the temperature of the wire is (Y = Young's modulus and s = specific heat capacity of the material of the wire).

(A) $\frac{(Mg)^2L}{2YAms}$

(B) $\frac{MgLms}{2YA}$

(C) $\frac{2YAL}{(Mg)^2ms}$

(D) none of these

Q.20

When a rubber cord is loaded, its length increases by 0.1%. If Poisson's ratio of rubber is 0.4, the percentage decrease in the diameter of the cord is

(A) 0.04%

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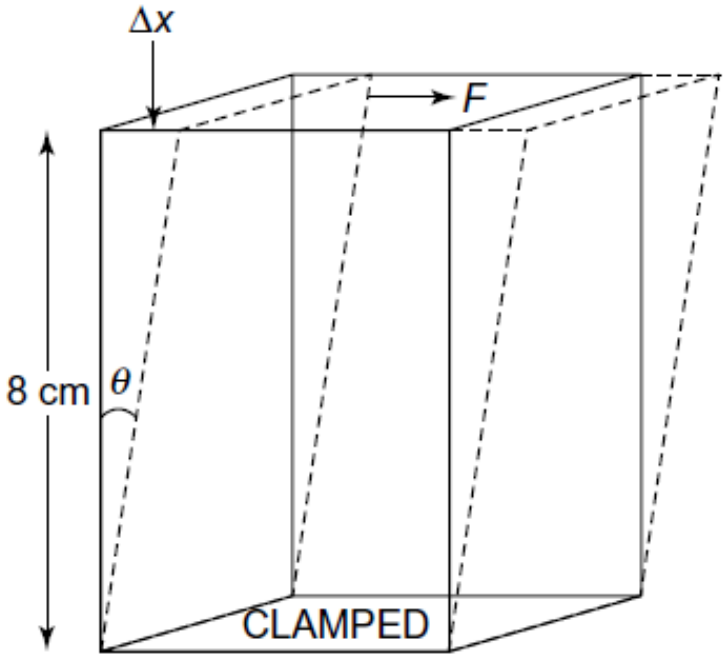
(B) 0.08%

(C) 0.10%

(D) 0.12%

Q.21

A rubber eraser $3\ \text{cm} \times 1\ \text{cm} \times 8\ \text{cm}$ is clamped at one end with $8\ \text{cm}$ edge vertical as shown in Fig. A horizontal force $F = 2.1\ \text{N}$ is applied on the free face. Calculate the horizontal displacement Δx of the top face. Given sheal modulus of rubber $= 1.4 \times 10^5 \text{Nm}^{-2}$ (in mm)



Q.22

A composite wire of a uniform cross-section of $5.5 \times 10^{-5}\ \text{m}^2$ consists of a steel wire of length $1.5\ \text{m}$ and a copper wire of length $2.0\ \text{m}$. By how much will it stretch when it is loaded with a mass of $200\ \text{kg}$? Young's modulus of steel is $2 \times 10^{11} \text{Nm}^{-2}$ and that of copper is $1 \times 10^{11} \text{Nm}^{-2}$. Take $g = 10\ \text{ms}^{-2}$ (in mm)

Q.23

Two spherical soap bubbles formed in vacuum have diameters $3.0\ \text{mm}$ and $4.0\ \text{mm}$. They coalesce to form a single spherical bubble. If the temperature remains unchanged, the diameter of the bubble so formed will be (in mm and round off to the nearest integer)

Q.24

If a rubber cube is taken $50\ \text{m}$ deep in a lake, its volume reduces by 0.5%. The bulk modulus of rubber is nearly (in Mega Pascal)

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Q.25

A wire suspended vertically from one of its ends is stretched by attaching a weight of $200\ \mathrm{N}$ to the lower end. The weight stretches the wire by $1\ \mathrm{mm}$. The elastic energy stored in the wire is (in mJ)

Answers & Solutions

Q.1 Answer:

$9.6 \times 10^{-2}\ \mathrm{m}$

Solution:

The weight of the rope can be assumed to act at its mid-point. Now, the extension l is proportional to original length L. If the weight of the rope acts at its mid-point, the extension will be that produced by half the rope. So, replacing L by $L/2$ in the expression for Young's modulus, we have

$$Y = \frac{FL/2}{Al} = \frac{FL}{2Al}$$

$$\therefore \quad l = \frac{FL}{2AY} = \frac{gL^2\rho}{2Y}$$

$(\because F = mg \text{ and } m = \rho V = \rho AL)$

Now $\rho = 1.5\ \mathrm{g\ cm^{-3}} = 1500\ \mathrm{kg\ m^{-3}}$, therefore

$$l = \frac{10 \times (8)^2 \times 1500}{2 \times 5 \times 10^6} = 9.6 \times 10^{-2}\ \mathrm{m}$$

Hence the correct choice is (a).

Q.2 Answer:

$1 : 8$

Solution:

Here $l_1 = \frac{FL_1}{\pi r_1^2 Y}$ and $l_2 = \frac{FL_2}{\pi r_2^2 Y}$. Therefore,

$$\frac{l_1}{l_2} = \frac{L_1}{L_2} \times \left(\frac{r_2}{r_1}\right)^2$$

Given $L_1 = 2L_1$ and $r_2 = r_1/2$. Thus $\frac{l_1}{l_2} = \frac{1}{2} \times \frac{1}{(2)^2} = \frac{1}{8}$. Hence the correct choice is (c).

Q.3 Answer:

$YA\alpha\theta$

Solution:

The coefficient of linear expansion is defined as

$$\alpha = \frac{\text{increase in length}}{\text{original length} \times \text{temp.rise}} = \frac{l}{L\theta}$$

$\therefore$ Increase in length $l = \alpha L\theta$. Now

$$Y = \frac{FL}{Al}$$

$$F = \frac{YAl}{L} = \frac{YA\alpha L\theta}{L} = YA\alpha\theta$$

Hence the correct choice is (c).

Q.4 Answer:

$1/2(F \times l)$

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Solution:

The force F can be assumed to act at the mid-point of the wire. Therefore, the average force responsible for extension is $F/2$. Thus, the work done by the force $\frac{F}{2}$ to produce an extension l in the wire = force $\times$ extension $= \frac{1}{2}(F \times l)$. Hence the correct choice is (b).

Q.5 Answer:

0.01 J

Solution:

We have seen above that work done by a force F to produce an extension l in the wire $= \frac{1}{2}F \times l$. Thus, the energy gained by the wire $= \frac{1}{2}F \times l = \frac{1}{2} \times 20 \times \left(1 \times 10^{-3}\right) = 10^{-2}$ J $= 0.01$ J. Hence the correct choice is (a).

Q.6 Answer:

0.075 J

Solution:

Energy stored per unit volume $= \frac{1}{2}Y \times \varepsilon^2$. Now

Strain $(\varepsilon) = \frac{l}{L} = \frac{1\,\text{mm}}{4\,\text{m}} = \frac{1 \times 10^{-3}\,\text{m}}{4\,\text{m}} = 2.5 \times 10^{-4}$

$\therefore$ Energy stored per unit volume $= \frac{1}{2} \times 2.0 \times 10^{11} \times \left(2.5 \times 10^{-4}\right)^2 = 6.25 \times 10^3 \text{Jm}^{-3}$

Volume of the wire $= \pi r^2 L = AL = 3 \times 10^{-6} \times 4 = 1.2 \times 10^{-5}$ m^3. Therefore, Energy stored in the wire

$= 6.25 \times 10^3 \times 1.2 \times 10^{-5} = 7.5 \times 10^{-2}$ J $= 0.075$ J

Hence the correct choice is (c).

Q.7 Answer:

length $= 0.5$ m, diameter $= 0.5$ mm

Solution:

We know that

$$F = \frac{YAl}{L} = \frac{Yl\pi r^2}{L}$$

$$l = \frac{FL}{\pi r^2 Y}$$

Since Y and F are the same for all wires, $l \propto \frac{L}{r^2}$ or $l \propto \frac{L}{d^2}$. It is easy to see that $\frac{L}{d^2}$ is the highest for the first wire. Hence the correct choice is (a).

Q.8 Answer:

1×10^5 N

Solution:

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The breaking strength is proportional to the square of the diameter $\left(F \propto d^2\right)$. Since the diameter is reduced to half, the breaking strength is reduced to onefourth. Hence the correct choice is (c).

Q.9 Answer:

1.25 mm

Solution:

When a wire of length L, cross-sectional area A and Young's modulus Y is stretched with a force F, the extension l in the wire is given by

$l = \frac{FL}{AY}$

Since F and A are the same for the two wires, we have

For copper wire $l_c = \frac{FL_c}{AY_c}$

For steel wire, $l_s = \frac{FL_s}{AY_s}$

$\therefore \quad l_s = l_c \frac{L_s}{L_c} \cdot \frac{Y_c}{Y_s}$

$= 1 \text{ mm} \times \left(\frac{0.5 \text{ m}}{1.0 \text{ m}}\right) \times \left(\frac{1.0 \times 10^{11} \text{Nm}^{-2}}{2.0 \times 10^{11} \text{Nm}^{-2}}\right)$

$= 0.25 \text{ mm}$

Hence the correct choice is (a).

Q.10 Answer:

$\sqrt{3} : \sqrt{2}$

Solution:

Young's modulus $Y = \frac{F}{A} \cdot \frac{L}{l}$

Force constant $k = \frac{F}{l} = \frac{YA}{L}$

where l is the extension in the spring of original length L and cross-sectional area A when a force $F = Mg$ is applied. Now, the time period of vertical oscillations is given by

$T = 2\pi\sqrt{\frac{M}{k}} = 2\pi\sqrt{\frac{ML}{YA}}$

$\therefore \frac{T_1}{T_2} = \sqrt{\frac{Y_2}{Y_1}} = \sqrt{\frac{3}{2}}$

Hence the correct choice is (a).

Q.11 Answer:

$\frac{3Mg}{2A}$

Solution:

Since the rod is uniform, half its weight can be taken to act at its mid-point. Therefore, stress at mid-point is weight of suspended mass + weight of half the rod cross-sectional area

$= \frac{Mg + \frac{1}{2}Mg}{A} = \frac{3Mg}{2A}$**, which is choice (b).**

Q.12 Answer:

$3 : 2$

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Solution:

Young's modulus $Y = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\varepsilon}$, **where** $\varepsilon = \frac{l}{L}$.
Now, if the temperature of a rod is increased by θ **the increase in its length due to thermal expansion is**

$$l = \alpha L \theta$$

$\therefore$ **Strain** $\varepsilon = \frac{l}{L} = \alpha\theta$. **Now stress is**

$$\sigma = Y\varepsilon = Y\alpha\theta$$

For the two rods, the stress is

$$\sigma_1 = Y_1\alpha_1\theta$$
$$\text{and} \quad \sigma_2 = Y_2\alpha_2\theta$$

But $\sigma_1 = \sigma_2$ **(given). Hence** $Y_1\alpha_1\theta = Y_2\alpha_2\theta$ **or** $\frac{Y_1}{Y_2} = \frac{\alpha_2}{\alpha_1} = \frac{3}{2}$. **Hence the correct choice is (c).**

Q.13 Answer:

$\frac{S^2}{2Y}$

Solution:

Let the strain be ε. **Then** $Y = \frac{S}{\varepsilon}$ **or** $\varepsilon = \frac{S}{Y}$. **Now energy density** $= \frac{1}{2}$ **stress** $\times$ **strain** $= \frac{1}{2} \times S \times \frac{S}{Y} = \frac{S^2}{2Y}$. **Hence the correct choice is (c).**

Q.14 Answer:

$\left(W_1 + \frac{3W}{4}\right)/S$

Solution:

Since the wire is uniform, i.e. its mass per unit length is constant over its entire length L, **the total downward weight** $=$ **the weight of the suspended mass** $+$ **weight of length** $\frac{3L}{4}$ **of the wire or** $F = W_1 + \frac{3W}{4}$

$$\therefore \quad \text{Stress} = \frac{\text{force}}{\text{area}} = \frac{F}{S} = \frac{W_1 + \frac{3}{4}W}{S}$$

Hence the correct choice is (c).

Q.15 Answer:

0.25×10^{-4}

Solution:

$\frac{\Delta L}{L} = \frac{\text{Stress}}{Y} = \frac{5\times10^7}{2\times10^{11}} = 2.5 \times 10^{-4}$. Given $\frac{\Delta V}{V}$ $= \frac{0.02}{100} = 2 \times 10^{-4}$

Now, $V = \pi r^2 L$. **Therefore,**

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$$\frac{\Delta V}{V}=\frac{\Delta\left(\pi r^2 L\right)}{\pi r^2 L}=\frac{2\pi r\Delta r L+\pi r^2 L\Delta L}{\pi r^2 L}$$
$$=2\frac{\Delta r}{r}+\frac{\Delta L}{L}$$
$$\text{or}\quad \frac{2\Delta r}{r}=\frac{\Delta V}{V}-\frac{\Delta L}{L}=2\times10^{-4}-2.5\times10^{-4}$$
$$=-0.5\times10^{-4}$$
$$\text{or}\quad \frac{\Delta r}{r}=-0.25\times10^{-4}$$

$\therefore$ Fractional decrease in $r=0.25\times10^{-4}$ which is choice (d).

Q.16 Answer:

$\frac{1}{2}ky(2x+y)$

Solution:

Potential energy stored in the spring when it is extended by x is

$U_1=\frac{1}{2}kx^2$

Potential energy stored in the spring when it is further extended by y is

$$U_2=\frac{1}{2}k(x+y)^2$$
$$\therefore\ \text{Work done}=U_2-U_1=\frac{1}{2}k(x+y)^2-\frac{1}{2}kx^2$$
$$=\frac{1}{2}ky(2x+y)$$

Hence the correct choice is (d).

Q.17 Answer:

2.5%

Solution:

Poisson's ratio is defined as

$\sigma=\frac{\Delta d/d}{\Delta l/l}\quad$ or $\quad\frac{\Delta l}{l}=\frac{\Delta d/d}{\sigma}$ **(i)**

where d is the diameter and l is the length of the wire. The area of cross-section is

$$A=\pi r^2=\frac{\pi d^2}{4}$$
$$\log A=\log\left(\frac{\pi}{4}\right)+2\log d$$

Differentiating, we have

$$\frac{\Delta A}{A}=2\frac{\Delta d}{d}$$
$$\frac{\Delta d}{d}=\frac{1}{2}\frac{\Delta A}{A}=\frac{1}{2}\times2\%=1\%$$
$$\left(\because\frac{\Delta A}{A}=2\%,\ \text{given}\ \right)$$

Using this in (i) we have

$\frac{\Delta l}{l}=\frac{1\%}{0.4}=2.5\%\quad(\because\sigma=0.4)$

Hence the correct choice is (b).

Q.18 Answer:

$\frac{mg}{AY\cos\theta}$

Solution:

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Refer to Fig. For vertical equilibrium

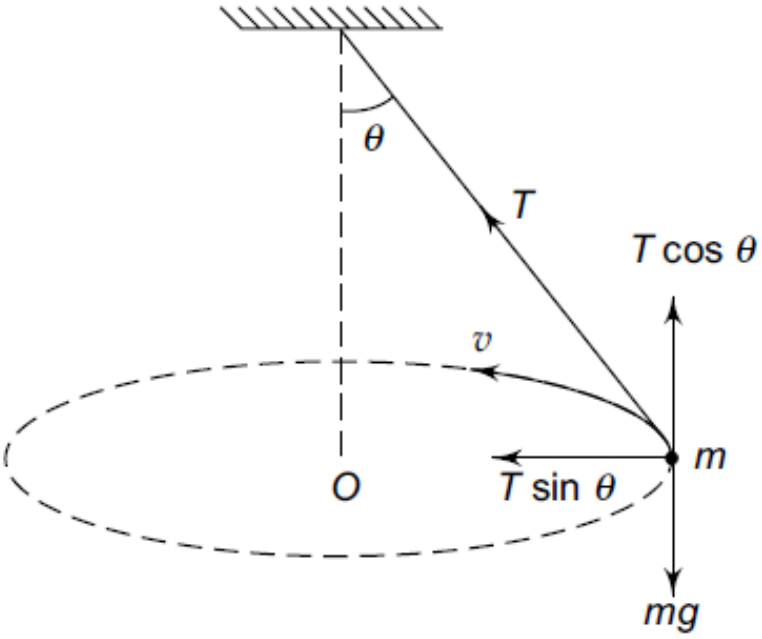
$T\cos\theta = mg$

or $T = \frac{mg}{\cos\theta}$

If L **is the original length of the wire, the increase in length is**

$$l = \frac{TL}{AY}$$

$$\therefore \quad \text{Strain} = \frac{l}{L} = \frac{T}{AY} = \frac{mg}{AY\cos\theta}$$



Q.19 Answer:

$\frac{(Mg)^2L}{2YAms}$

Solution:

Stress $= \frac{Mg}{A}$**, strain** $= \frac{\Delta L}{L}$**, volume** $= AL$

Energy stored in wire is

$U =$ **energy stored per unit volume** $\times$ **volume**

$= \left(\frac{1}{2}\right.$ **stress** $\times$ **strain** $\left.\right) \times$ **volume**

$= \frac{1}{2} \times \left(\frac{Mg}{A} \times \frac{\Delta L}{L}\right) \times AL$

$= \frac{1}{2}Mg\Delta L$

$= \frac{(Mg)^2L}{2AY} \quad \left(\because \Delta L = \frac{MgL}{AY}\right)$

If ΔT **is the rise in temperature, then** $U = ms\Delta T$**, i.e.**

$$\frac{(Mg)^2L}{2AY} = ms\Delta T$$

$$\Rightarrow \quad \Delta T = \frac{(Mg)^2L}{2AYms}, \text{ which is choice (a)}$$

Q.20 Answer:

0.04%

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Solution:

Poisson's ratio $= \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$

or $\sigma = -\frac{\Delta D/D}{\Delta L/L}$

$$\therefore \quad \left|\frac{\Delta D}{D}\right| = \sigma \times \frac{\Delta L}{L} = 0.4 \times 0.1\%$$

$$= 0.04\%$$

$\therefore$ So the correct choice is (a).

Q.21 Answer:

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Solution:

$\tan\theta = \frac{\Delta x}{8\text{ cm}} \Rightarrow \theta = \frac{\Delta x}{8\text{ cm}}$ **($\because \theta$ is small)**

$\therefore \quad \Delta x = 8\text{ cm} \times 0.05 = 0.4\text{ cm} = 4\text{ mm}$

Q.22 Answer:

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Solution:

Since the steel and copper parts of the wire are of the same cross-sectional area (A) and are loaded with the same weight $F = Mg$, they are under the same stress (F/A). Therefore,

$\frac{Y_s l_s}{L_s} = \frac{Y_c l_c}{L_c} = \frac{F}{A} = \frac{Mg}{A}$

or $l_s = \frac{Mg}{A} \cdot \frac{L_s}{Y_s}$ **and** $l_c = \frac{Mg}{A} \cdot \frac{L_c}{Y_c}$

$\therefore$ Extension of the composite wire is

$l = l_s + l_c = \frac{Mg}{A}\left(\frac{L_s}{Y_s} + \frac{L_c}{Y_c}\right)$

Substituting the given values, we find that $l = 10^{-3}\text{ m} = 1\text{ mm}.$

Q.23 Answer:

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Solution:

Since the bubbles are in vacuum, the pressure of air inside them are $P_1 = \frac{4\sigma}{r_1}$ **and** $P_2 = \frac{4\sigma}{r_2}$**, where** $r_1 = 3.0\text{ mm}$ **and** $r_2 = 4.0\text{ mm}$**. Since the temperature remains unchanged, we have from Boyle's law**

$$P_1V_1 + P_2V_2 = PV$$

$$\text{or } \frac{4\sigma}{r_1} \cdot \frac{4}{3}\pi r_1^3 + \frac{4\sigma}{r_2} \cdot \frac{4}{3}\pi r_2^3 = \frac{4\sigma}{r} \cdot \frac{4}{3}\pi r^3 \quad \text{(i)}$$

where r is the radius of the single bubble formed. From (i), we get $r^2 = r_1^2 + r_2^2$ **or** $r = \sqrt{r_1^2 + r_2^2}$

$= \sqrt{(3.0)^2 + (4.0)^2} = 5.0\text{ mm}$

Q.24 Answer:

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Solution:

Excess pressure at depth $h = 50\ \text{m}$ **is**

$P = h\rho g = 50 \times 1000 \times 9.8 = 4.9 \times 10^5\ \text{Pa}$

Volume strain $\frac{\Delta V}{V} = -0.5\% = -0.5 \times 10^{-2}$

$\therefore$ **Bulk modulus** $= -\frac{P}{\frac{\Delta V}{V}} = -\frac{4.9\times10^5}{-0.5\times10^{-2}} \simeq 10^8\ \text{Pa}$

Q.25 Answer:

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Solution:

The work done by a force in stretching a wire is stored in the wire as potential energy. The work done by a force F **in stretching the wire by a small amount** dl **is**

$dW = Fdl$

The work done in stretching a wire by l **is**

$W = \int_0^l Fdl$

Now $F = \frac{YAl}{L}$**. Therefore**

$$W = \frac{YA}{L}\int_0^l ldl = \frac{YAl^2}{2L} = \frac{1}{2}Fl \left(\because F = \frac{YAl}{L}\right)$$

$$= \frac{1}{2} \times 200 \times \left(1 \times 10^{-3}\right)$$

$$= 0.1\ \text{J}$$

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